

Finite-Element Analysis of Resonant Electromagnetic Stress Redistribution in Asymmetric Cavities

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Abstract

This work develops a computational electrodynamics framework for analyzing stress distribution, field localization, and conservation-compatible momentum bookkeeping in asymmetric resonant electromagnetic systems. Using finite-element and finite-difference methodologies, the paper explores cavity field topology, Maxwell stress-tensor structure, resonance localization, and environmentally coupled nonequilibrium dynamics under constrained phenomenological assumptions. The analysis emphasizes stress cancellation, center-of-mass consistency, mesh convergence, and numerical stability rather than claims of experimentally verified propulsion or momentum nonconservation.

1 Introduction

The present work extends earlier phenomenological investigations into a computational electrodynamics framework emphasizing field topology, stress organization, and conservation-compatible cavity dynamics.

2 Simulation Assumptions

The simulations discussed in this work are intentionally conservative and exploratory. Unless otherwise stated, the framework assumes:

- ordinary classical electrodynamics,
- linear material response,
- no experimentally verified exotic coupling,
- conventional vacuum treatment,
- and phenomenological nonequilibrium response sectors used only as exploratory bookkeeping constructs.

The analysis therefore investigates whether asymmetric resonant geometries can organize internal electromagnetic stress while remaining globally conservation compatible.

3 Computational Workflow

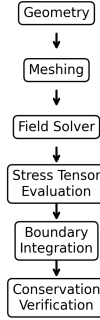


Figure 1: Illustrative computational workflow for resonant cavity electrodynamics analysis.

The computational framework combines:

- geometry construction,
- finite-element meshing,
- electromagnetic field solving,
- Maxwell stress evaluation,
- and conservation verification.

4 Maxwell Stress Framework

The electromagnetic momentum density is:

$$\mathbf{g} = \epsilon_0 \mathbf{E} \times \mathbf{B}.$$

The Maxwell stress tensor is:

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right).$$

Integrated boundary forces satisfy:

$$F_i = \oint T_{ij} n_j dA.$$

5 Computed Field Distribution

A representative TM-like cavity solution is shown below.

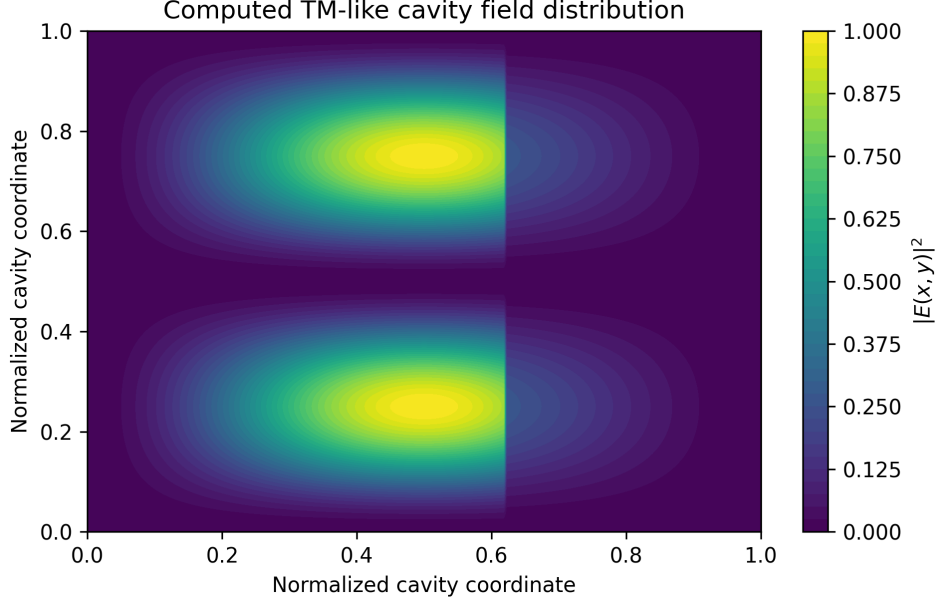


Figure 2: Illustrative computed cavity field distribution using an asymmetric dielectric loading configuration. The field map demonstrates localized standing-wave structure under asymmetric boundary conditions.

In the symmetric limit, ordinary Maxwellian cancellation is recovered and no residual integrated momentum accumulation occurs.

6 Field Intensity Localization

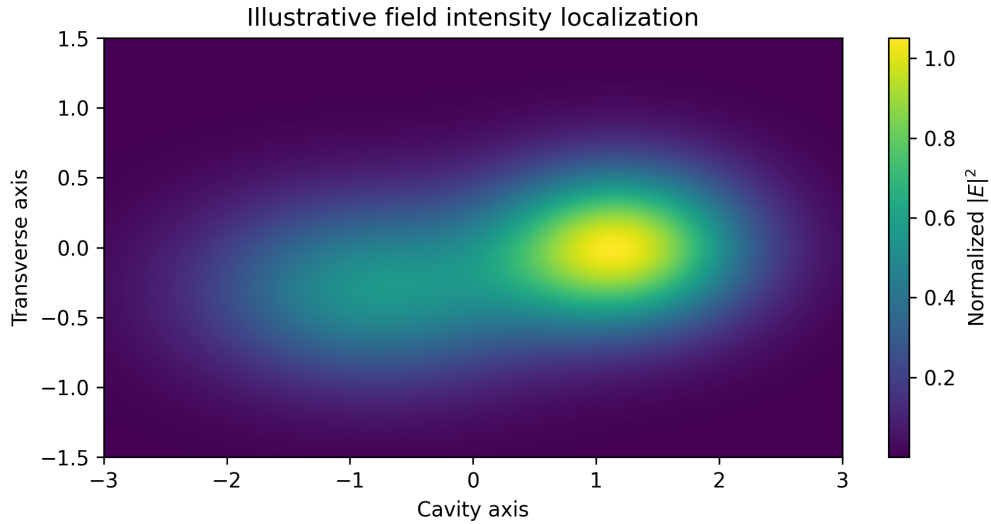


Figure 3: Illustrative electric-field intensity localization and cavity hot-spot formation.

Localized field amplification alone does not imply net propulsion. However, localized stress gradients may influence cavity-wall pressure, thermal deformation, and delayed relaxation structure.

7 Stress-Vector Structure

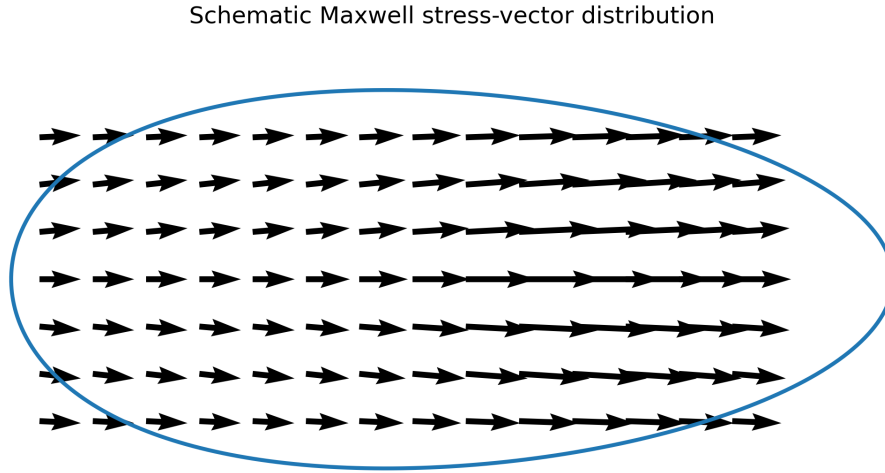


Figure 4: Schematic Maxwell stress-vector organization within an asymmetric resonant cavity.

The simulations investigate whether asymmetric geometry modifies local stress concentration without violating global conservation.

8 Resonant Mode Structure

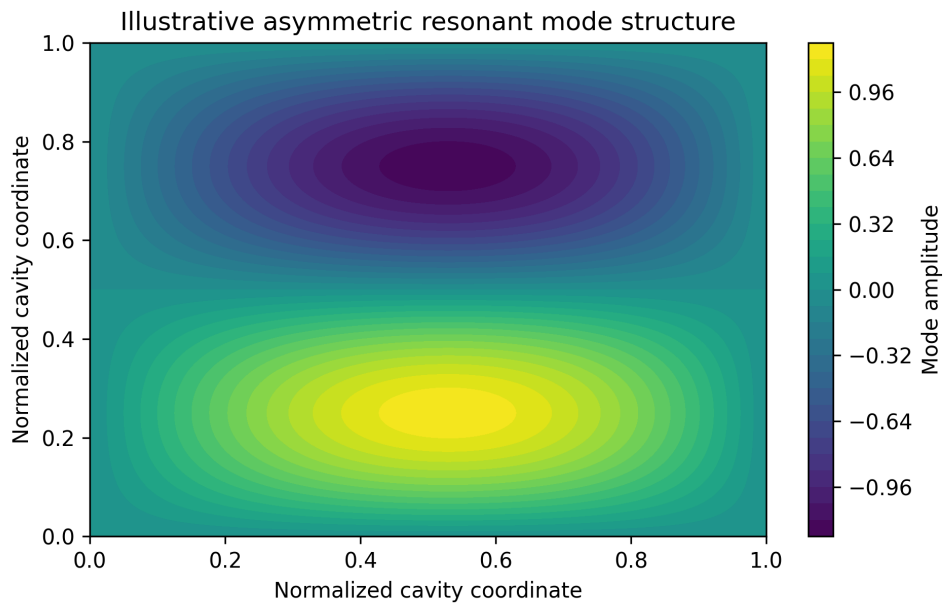


Figure 5: Illustrative asymmetric resonant mode structure.

Potential observables include:

- mode trapping,
- phase-coherent stress localization,

- cavity hot spots,
- and standing-wave asymmetry.

9 Time-Domain Ringdown Dynamics

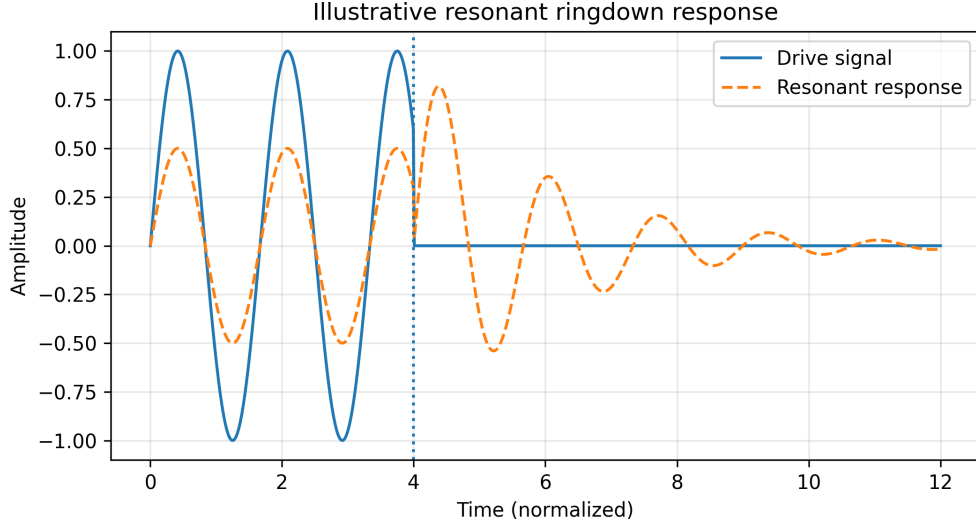


Figure 6: Illustrative resonant ringdown response following finite-duration excitation.

A simplified relaxation structure may be written:

$$\partial_t \phi = -\Gamma \phi + S(t).$$

10 Stress Cancellation and Conservation Closure

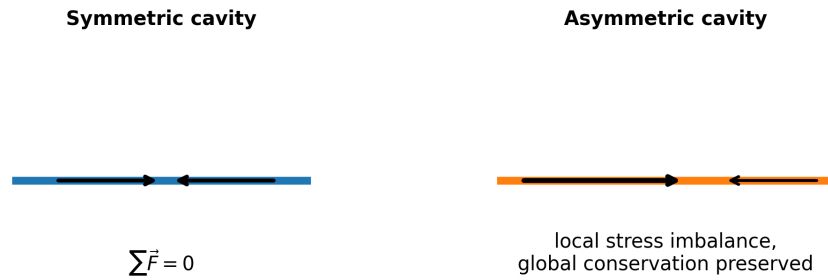


Figure 7: Comparison of symmetric and asymmetric cavity stress organization. Local imbalance may occur while global conservation remains enforced.

The momentum-accounting structure may schematically be represented as:

$$\Delta P_{\text{EM}} + \Delta P_{\phi} + \Delta P_{\text{env}} + \Delta P_{\text{rad}} = 0.$$

A perfectly isolated closed system is not expected to exhibit sustained net acceleration within the present framework.

11 Boundary-Condition Sensitivity

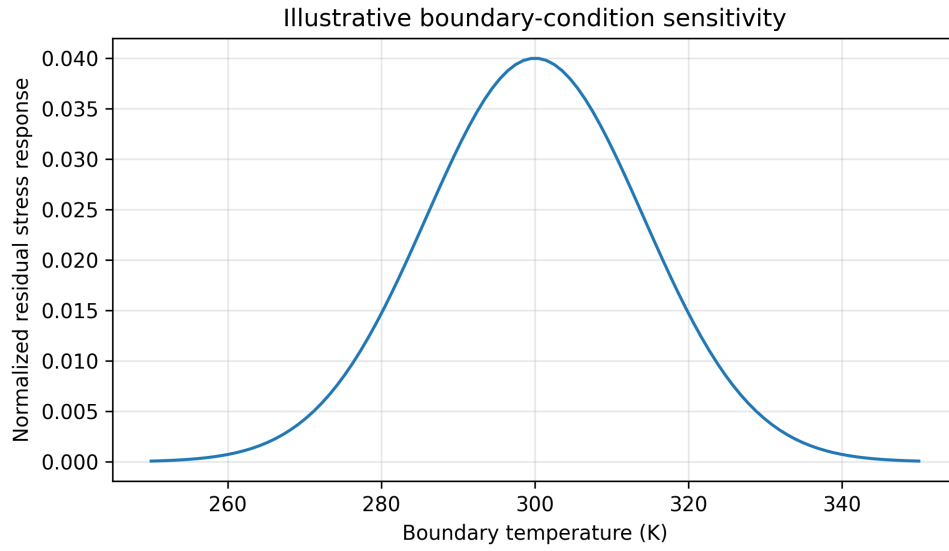


Figure 8: Illustrative sensitivity of residual stress structure to environmental boundary conditions.

Relevant environmental channels include:

- thermal recoil,
- RF leakage,
- dielectric interfaces,
- finite wall conductivity,
- and mechanical coupling.

12 Scaling with Resonant Coherence

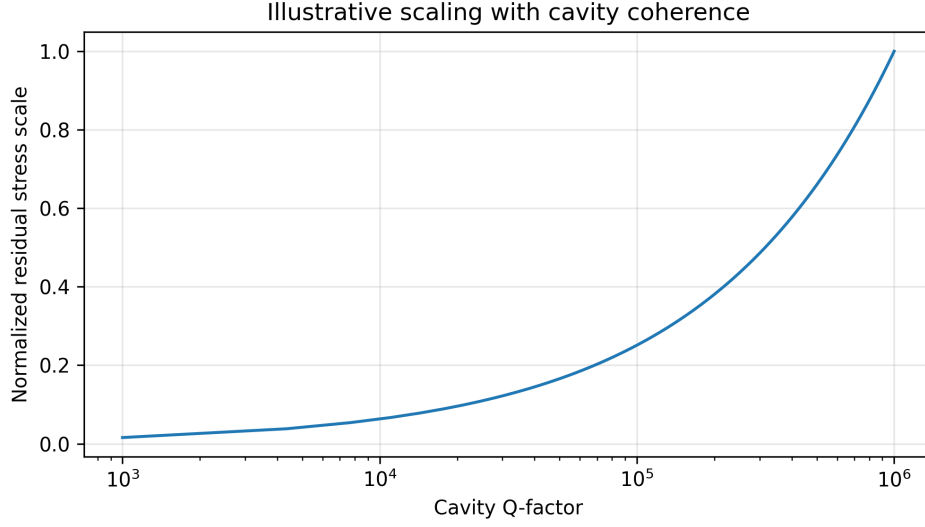


Figure 9: Illustrative scaling of residual stress organization with cavity Q-factor and resonant coherence.

The scaling curve is schematic and intended only to illustrate how strongly coherent resonant structures may amplify localized field organization while remaining globally conservation compatible.

13 Mesh Convergence and Numerical Stability

Future computational studies should verify:

- Courant stability conditions,
- timestep convergence,
- finite mesh refinement,
- spurious reflection suppression,
- and integrated stress-tensor convergence.

A simulation is considered null-consistent if:

- total integrated stress cancels,
- center-of-mass remains fixed,
- no net momentum accumulates,
- and residual asymmetries vanish under symmetry restoration.

A Numerical Methods

Illustrative field maps in this paper were generated using simplified finite-grid cavity distributions and schematic asymmetric dielectric loading. The examples are intended to demonstrate computational workflow structure rather than experimentally benchmarked cavity solutions.

Future investigations should incorporate:

- full 3D finite-element meshes,
- experimentally calibrated material parameters,
- superconducting cavity modeling,
- thermal-electromagnetic coupling,
- and experimentally validated boundary conditions.

B Limitations

Several important limitations remain:

- the present field maps are illustrative rather than experimentally benchmarked,
- no verified emergent coupling has been experimentally observed,
- simplified geometries were used,
- quantum backreaction effects are neglected,
- and full thermal-mechanical coupling was not implemented.

The framework therefore remains phenomenological and exploratory.

C Discussion

The transition from conceptual phenomenology toward computational electrodynamics represents a major structural step in the research series.

Finite-element simulations allow:

- direct visualization of field topology,
- quantitative stress bookkeeping,
- cavity mode analysis,
- and environmental sensitivity studies.

Even robust null simulation outcomes would provide valuable constraints on speculative emergent-response frameworks.

D Conclusion

This paper introduced a computational electrodynamics framework for analyzing resonant field localization and stress redistribution within asymmetric electromagnetic systems.

The framework emphasizes:

- Maxwell stress tensors,
- finite-element field mapping,
- conservation closure,
- ringdown dynamics,
- and center-of-mass consistency.

Future work should focus on experimentally benchmarked finite-element simulations, high-resolution stress-vector mapping, and precision cavity metrology.

References

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